A Real-World Multi-modal Dataset for Surveillance Under High Environmental Variations *(Supplementary Document)*

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Keywords

multi-modal dataset, visual surveillance, adverse weather condition, vision language model.

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1 Dataset:

The video samples of HEV-Bench can be accessed through this github link: HEV-Bench.

2 Supplementary images:

This additional document represents more samples of each of the environmental condition-**Haze, Night and Rain** with three categories: **Occlusion, Small Target and Occlusion + Small Target**. A few video samples and annotations are mentioned in Fig. 4 and Fig. 5. Some more results are shown in Fig. 6.

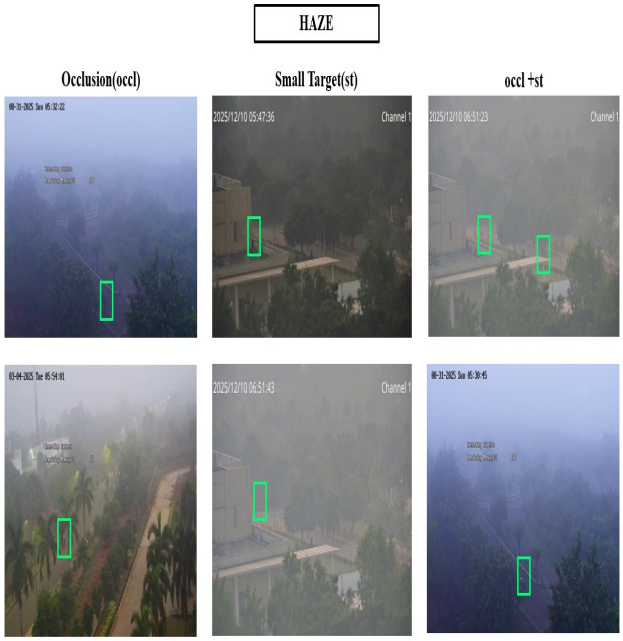

Figure 1: These images belong to hazy environment. Here, the green bounding boxes represent the objects.

References

[1] Daya Guo et al. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948 [cs.CL]
[2] Gemini and Team. 2025. Gemini: A Family of Highly Capable Multimodal Models. arXiv:2312.11805 [cs.CL] https://arxiv.org/abs/2312.11805
[3] OpenAI and Team. 2024. GPT-4 Technical Report. arXiv:2303.08774 [cs.CL]
[4] Hang Zhang, Xin Li, and Lidong Bing. 2023. Video-LLaMA: An Instruction-tuned Audio-Visual Language Model for Video Understanding. *arXiv preprint arXiv:2306.02858* (2023).

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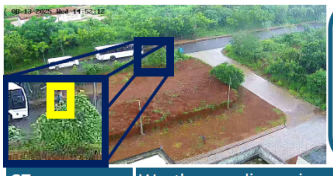

PROMPT

Describe the weather and any object movement. If there is movement, specify the direction. Describe it in a few lines.

GT	Weather: medium rain Movement: A person can be seen moving towards the bus from right to left in the screen
GPT[3]	weather: clear and dry conditions, no rain or fog movement: a person or vehicle movement is observed, moving from right to left across the scene
DeepSeek[1]	weather: clear with bright daylight. movement: a white vehicle continues moving away from the camera, traveling further into the distance down the road. no person movement is observed.
GEMINI[2]	weather: it is raining and overcast with wet roads and ground. movement: a person is walking along the road toward the left, and a dark car appears at the end of the footage driving toward the right.
Video-Llama[4]	First, we see a scene with clear and dry weather conditions. The visibility is good, and there are no signs of rain, fog, or other atmospheric disturbances. Next, the camera maintains a steady view of the area under these stable conditions. The environment appears calm and unchanged. Then, movement is observed as a person or a vehicle passes through the scene, traveling from right to left across the frame.

Figure 4: Here, the green bounding boxes represent the objects and the blue bounding boxes represent the enlarged images. The descriptions are generated using [2], [3], [1], [4].

PROMPT

Describe the weather and any object movement. If there is movement, specify the direction. Describe it in a few lines.

GT	Weather: Early morning with heavy fog/mist, low visibility. no rainfall visible. Movement: A person can be seen walking on the footpath from east to west near the longitudinally-mid part of the screen. The cyclist can be seen moving from west to east on the pathway often getting hidden behind the canopies. The two cross each other. They carry past each other in opposite direction.
GPT[3]	Weather: Overcast conditions with dull, even lighting. No rain, fog, or visible wind effects. Movement: No noticeable movement detected in this clip.
DeepSeek[1]	Weather: Overcast, uniform cloud cover. None visible. Unobstructed, no fog or mist. Movement: The scene is static across all five frames. No objects, vehicles, or environmental changes are observed.
GEMINI[2]	Weather: The weather appears to be hazy or foggy, significantly reducing visibility in the distance. Movement: There is very minimal movement detected. A faint, dark shape, possibly a person, is seen walking away from the camera on the path in the middle-right of the frame, moving right-to-left.
Video-Llama[4]	First, the video shows a shot of a house with a tree in front of it. The sky is gray, and the air is foggy, which suggests a heavy smog or pollution in the area. There is also a large building in the distance, which is surrounded by trees. Next, the camera moves to a tree-covered hill, which is visible from the street. The sky is still gray and misty, and there is a sign near a fence in the distance. Then, the video shows an aerial view of a city with many buildings and trees in the background. The buildings are gray, and there are buildings in the distance with green trees. The sky is still gray, and the air is heavy with smog. Finally, the video captures a shot of a tree in front of a large building. There are many trees in the distance, and the sky is still gray. Overall, the video portrays a hazy and polluted atmosphere that is harmful to people and their health.

Figure 5: Here, the green bounding boxes represent the objects and the blue bounding boxes represent the enlarged image. The descriptions are generated using [2], [3], [1], [4].

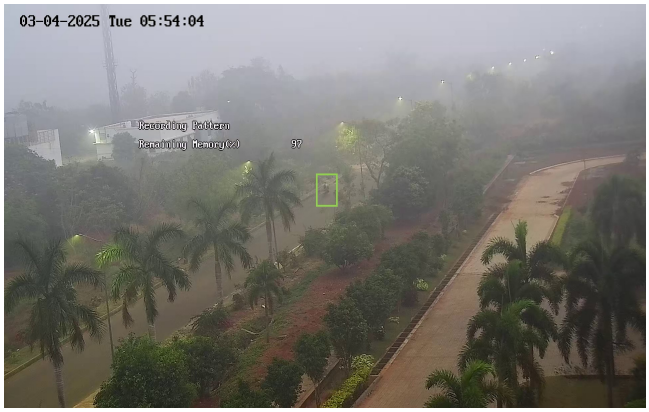

Figure 6: The green bounding boxes represent the objects. Here, the bike is entering into dense haze.

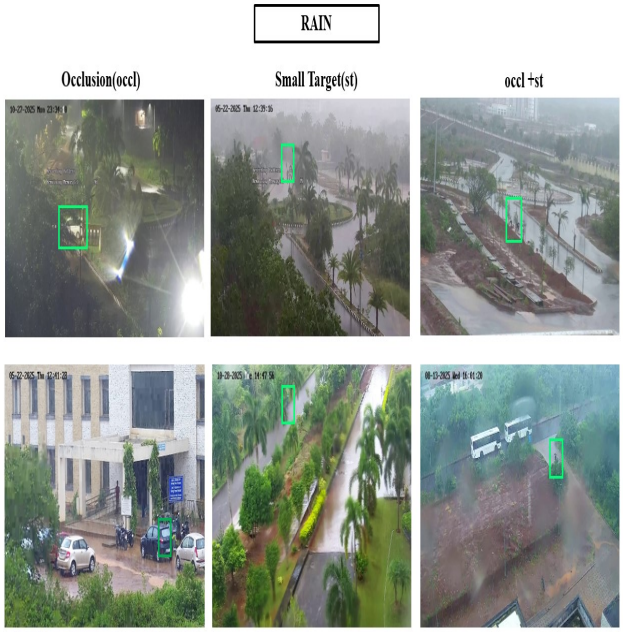

Figure 2: Here, the green bounding boxes represent the objects and the blue bounding boxes represent the enlarged image.

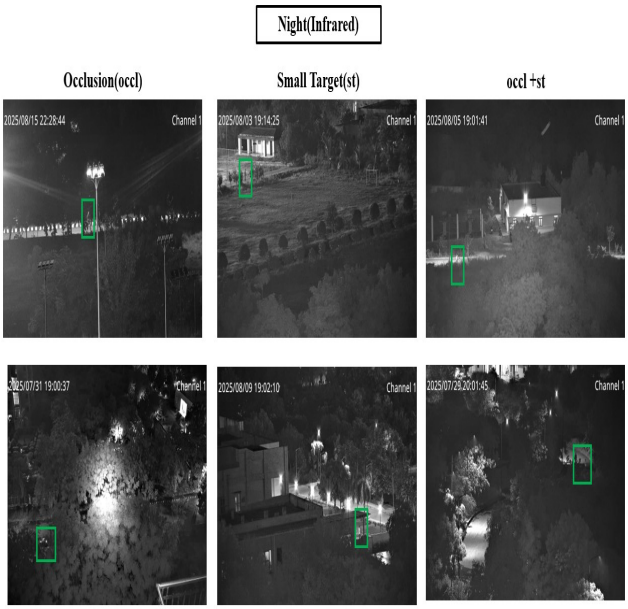

Figure 3: These images belong to Night environment. Here, the green bounding boxes represent the objects.